

EENG250 ELECTRIC CIRCUITS I · CHAPTER 1

Circuit Variables

Four quantities and one sign rule you will use in every chapter: charge, current, voltage and power.

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PREDICT FIRST

Your home has a 5 A generator subscription. Can you run a 2000 W kettle?

5 A × 220 V ≈ 1.1 kW. No: the breaker trips. The ampere size caps your **power** at any moment; the kWh meter counts your **energy** over the month.

TODAY

By the end of this session you can

1 • Units

Use SI units and the prefixes p, n, μ , m, k, M, G.

2 • Current and voltage

$$i = \frac{dq}{dt}, \quad v = \frac{dw}{dq}$$

3 • Power and energy

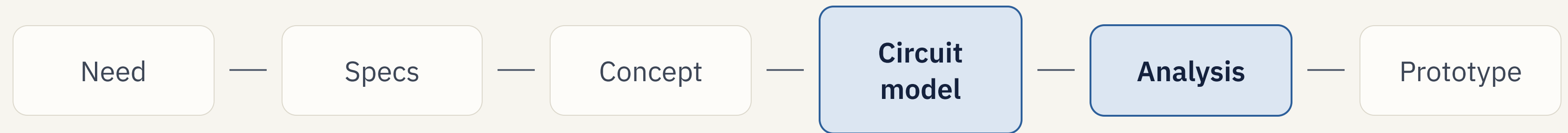
$$p = vi, \quad w = \int p dt$$

4 • The sign rule

Say from a diagram whether an element absorbs or delivers power.

CIRCUITS ARE MODELS

A circuit is a mathematical model of a real system



Measurements on the prototype send you back to refine the model. **This course is the two blue boxes:** we predict a circuit's behaviour from its voltages and currents before anyone builds it.

WHEN CIRCUIT THEORY APPLIES

The system must be small compared with the wavelength

$$\lambda = \frac{c}{f}, \quad c = 3 \times 10^8 \text{ m/s}$$

Nilsson's one-tenth rule: the system is **lumped** if it is smaller than one tenth of λ .

Signal	f	Wavelength λ	Lumped if smaller than
Lebanese power grid	50 Hz	6000 km	600 km
FM radio	100 MHz	3 m	30 cm
1 GHz radio link	1 GHz	30 cm	3 cm

Three assumptions follow: effects are instantaneous, net charge on each component is zero, no magnetic coupling between components.

SI UNITS

Six units carry this whole course

Quantity	Unit	Symbol	Equals
Charge	coulomb	C	$A \cdot s$
Current	ampere	A	C/s
Voltage	volt	V	J/C
Power	watt	W	$J/s = V \cdot A$
Energy	joule	J	$N \cdot m = W \cdot s$
Resistance	ohm	Ω	V/A

Units cost real missions: NASA lost the Mars Climate Orbiter in 1999 when pound-force seconds met software that expected newton-seconds.

PREFIXES

Pick the prefix that puts the number between 1 and 1000

p

10^{-12}

n

10^{-9}

μ

10^{-6}

m

10^{-3}

k

10^3

M

10^6

G

10^9

0.00001 s → **10 μs**, not 0.01 ms or 10,000 ns.

Your turn: 2,200,000 W = ? 0.000 330 A = ? 0.000 000 047 s = ?

CURRENT

Current is the rate of charge flow

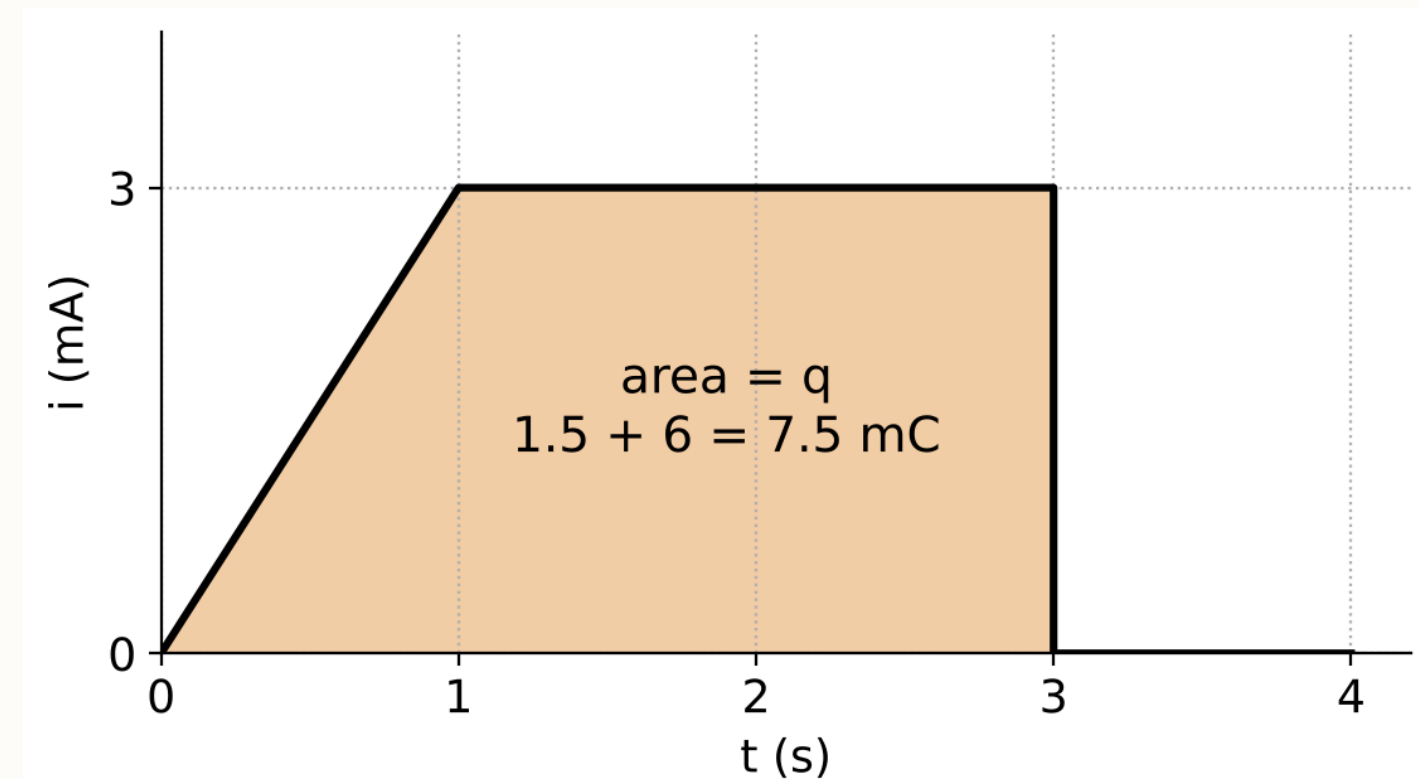
$$i = \frac{dq}{dt}$$

$$q(t) = \int_0^t i(x) dx$$

1 A = 1 C/s. Current is the **slope** of $q(t)$; charge is the **area** under $i(t)$, taking $q = 0$ at $t = 0$.

Graph: $\frac{1}{2}(1 \text{ s})(3 \text{ mA}) + (2 \text{ s})(3 \text{ mA}) = 1.5 + 6 = 7.5 \text{ mC}$.

Nilsson Example 1.2: 5 A for 10 s delivers 50 C.



VOLTAGE

Voltage is energy per unit charge

$$v = \frac{dw}{dq} \qquad 1 \text{ V} = 1 \frac{\text{J}}{\text{C}}$$

Separation of charge

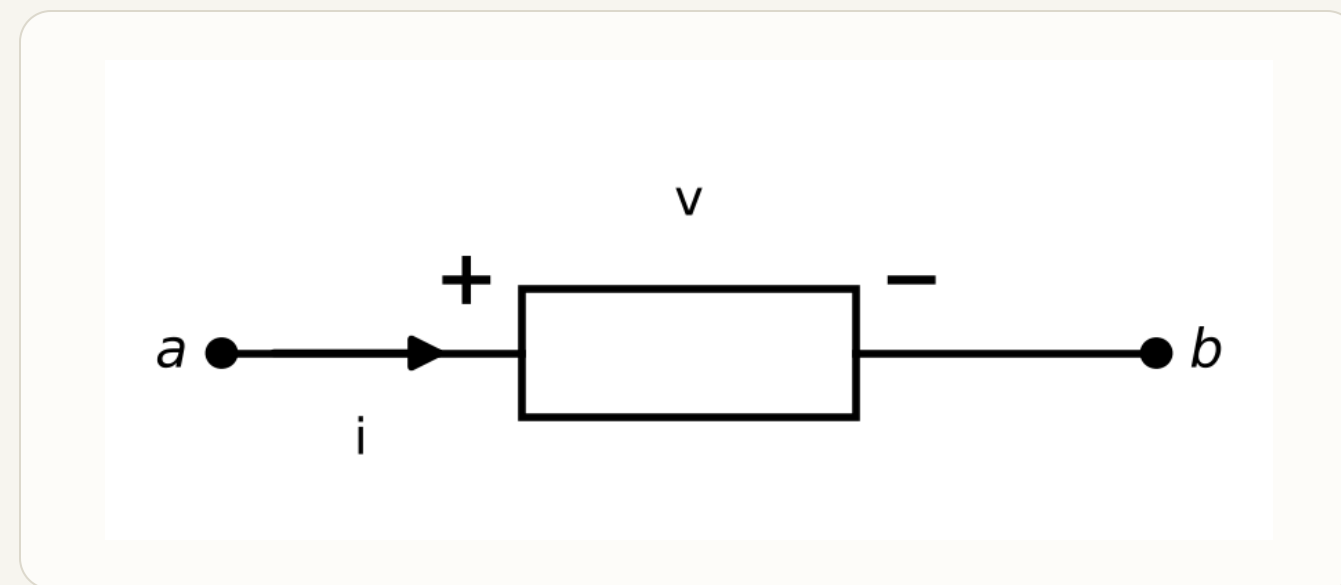
A 12 V battery gives 12 J to every coulomb that passes through it from – to +.

Always between two points

"The voltage at a terminal" means nothing until you say: relative to which other point?

REFERENCE DIRECTIONS

The + / – and the arrow are labels; the sign of the number is the physics



Value

v

i

Positive

a is higher than b

+ charge moves along the arrow

Negative

b is higher than a

+ charge moves against the arrow

You choose the labels freely. Once chosen, every equation must agree with them. Electrons move opposite to the positive-charge flow.

BUILDING BLOCKS

An ideal basic circuit element has three attributes

1

It has only two terminals.

2

It is described by its voltage and current.

3

It cannot be subdivided into other elements.

Ideal = a model, not a part you can buy. Real devices are built from combinations of ideal elements.

POWER

Power is the product of voltage and current

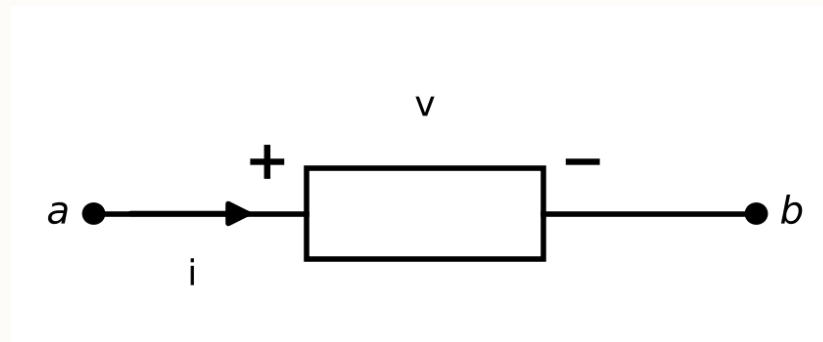
$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

1 W = 1 J/s. Power is a property of **one element**: the voltage across it times the current through it.

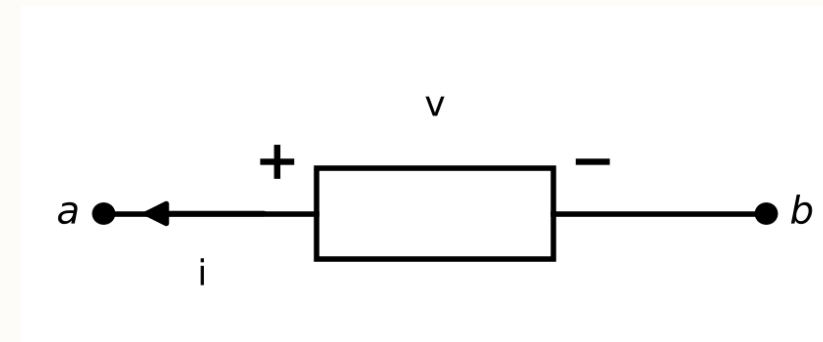
The useful output is often non-electrical (light, sound, heat, motion), and every real device has a power limit. That is why designs are specified in watts.

THE PASSIVE SIGN CONVENTION

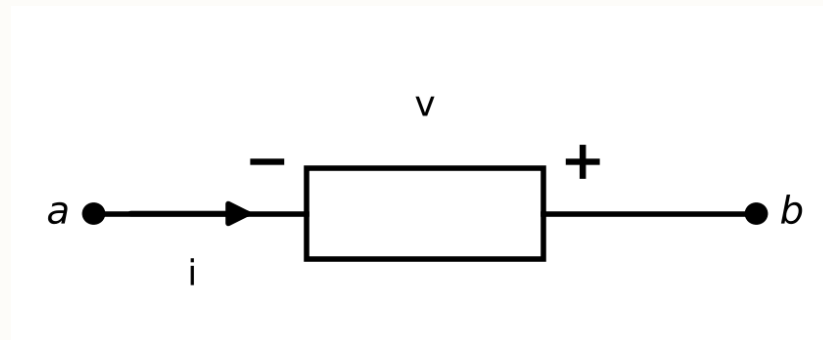
Arrow enters the + terminal: $p = vi$. Otherwise: $p = -vi$



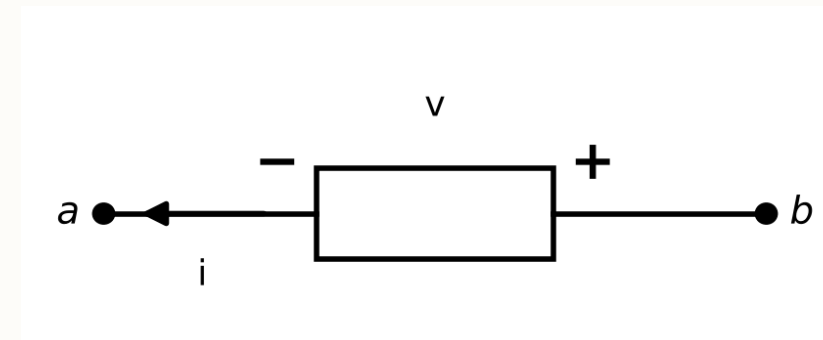
(a) $p = vi$



(b) $p = -vi$



(c) $p = -vi$



(d) $p = vi$

Find the terminal where the current **enters the element**. An arrow pointing away from one terminal means the current enters at the other.

READING THE SIGN

$p > 0$: the element absorbs. $p < 0$: it delivers.

Nilsson Example 1.3: two students analyse the same element with different labels.

Student 1, labels (b)

$$i = 4 \text{ A}, v = -10 \text{ V}$$

$$p = -vi = -(-10)(4) = \mathbf{40 \text{ W, absorbs}}$$

Student 2, labels (c)

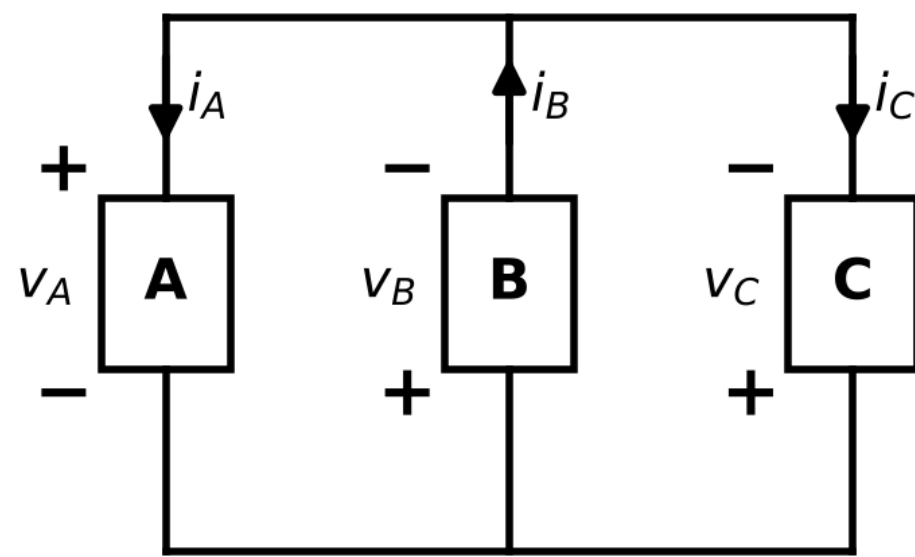
$$i = -4 \text{ A}, v = 10 \text{ V}$$

$$p = -vi = -(10)(-4) = \mathbf{40 \text{ W, absorbs}}$$

Different labels, different numbers, **the same physics.**

PREDICT FIRST · 3 MINUTES, NO CALCULATOR

Which elements absorb power?



Element

v (V)

i (A)

A

10

2

B

-10

5

C

-10

3

1. Does each arrow enter + or -?
2. Absorbs or delivers?
3. Do the three powers add to zero?

ANSWER

A and C absorb; B delivers 50 W

Element	Arrow enters	Power	Result
A	+ (top)	$(10)(2) = 20 \text{ W}$	absorbs
B	+ (bottom): the arrow points up, out of the top	$(-10)(5) = -50 \text{ W}$	delivers
C	– (top)	$-(-10)(3) = 30 \text{ W}$	absorbs

Check: $20 - 50 + 30 = 0$. Delivered 50 W = absorbed 50 W.

ENERGY

Energy is power accumulated over time

$$w(t) = \int_0^t p(x) dx \quad \text{constant } P: \quad w = P t$$

The kWh on your bill

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \text{ MJ}$$

A 2000 W kettle for 3 min

$$2000 \text{ W} \times 180 \text{ s} = 360 \text{ kJ} = 0.1 \text{ kWh}$$

Time in **seconds** gives joules. Hours or minutes are the commonest unit slip.

Power in an element whose voltage and current both decay

Given. $v = 0$ and $i = 0$ for $t < 0$. For $t \geq 0$, with the current entering the + terminal (so $p = +vi$):

$$v = 10 e^{-5000t} \text{ kV} = 10\,000 e^{-5000t} \text{ V}, \quad i = 20 e^{-5000t} \text{ A}$$

Step 1 • multiply

$$p = vi = (10\,000 e^{-5000t}) (20 e^{-5000t}) = 200\,000 e^{-10\,000t} \text{ W}$$

$$e^a e^b = e^{a+b}: \quad -5000t - 5000t = -10\,000t$$

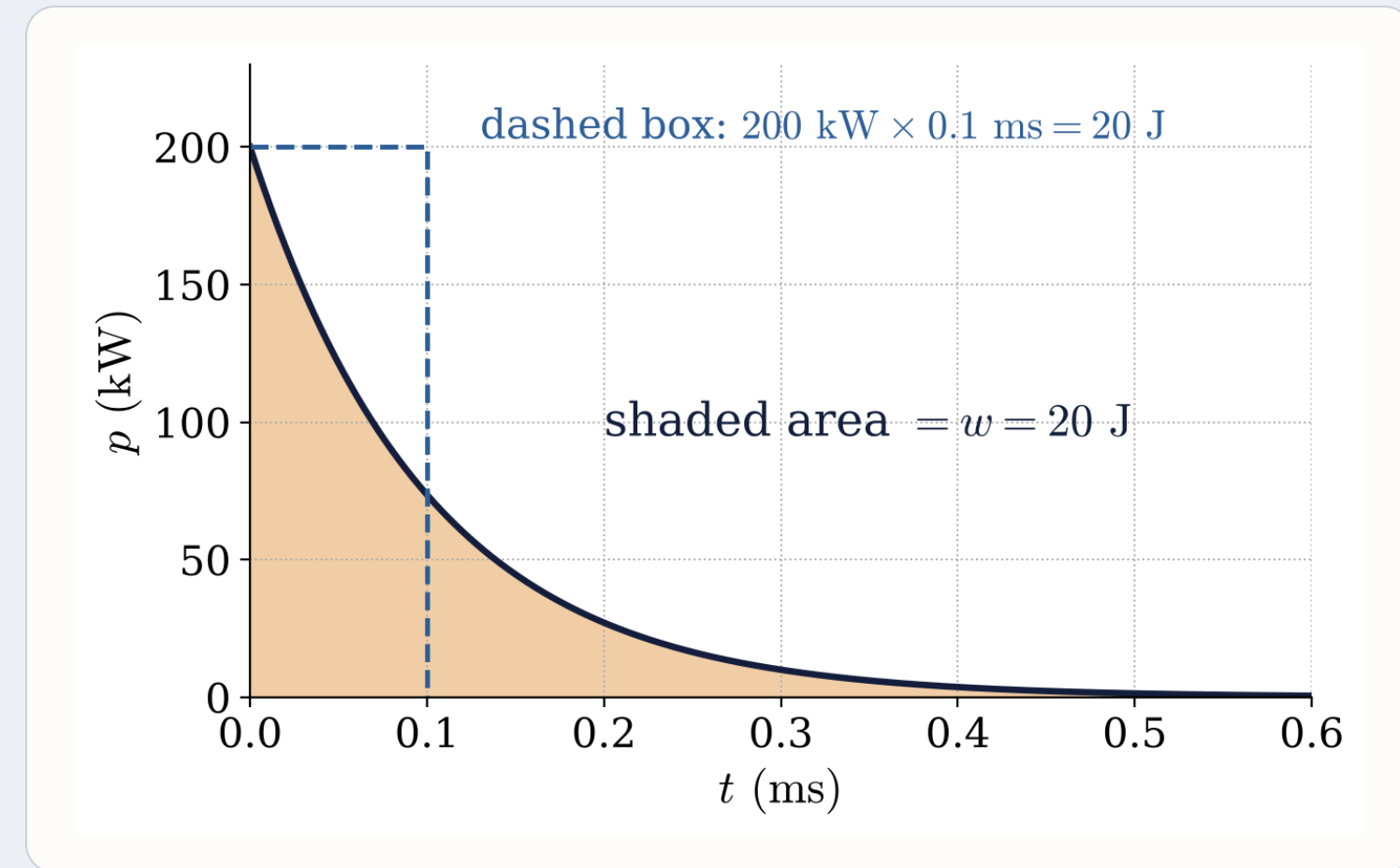
Step 2 • at 1 ms

$$t = 1 \text{ ms}: \quad -10\,000 \times 0.001 = -10$$

$$p(1 \text{ ms}) = 200\,000 e^{-10} = 200\,000 \times 4.54 \times 10^{-5} = 9.08 \text{ W}$$

Convert kV to V before multiplying: $10 \text{ kV} \times 20 \text{ A}$ is $10,000 \text{ V} \times 20 \text{ A} = 200,000 \text{ W}$ at $t = 0$.

The energy delivered is the area under $p(t)$



$$\begin{aligned}
 w &= \int_0^{\infty} p \, dt = \int_0^{\infty} 200\,000 e^{-10\,000t} \, dt \\
 &= \left[-\frac{200\,000}{10\,000} e^{-10\,000t} \right]_0^{\infty} = \left[-20 e^{-10\,000t} \right]_0^{\infty} \\
 &= 0 - (-20) = 20 \text{ J}
 \end{aligned}$$

Check. The area under a decaying exponential equals its starting value times its time constant, here $1/10,000 \text{ s} = 0.1 \text{ ms}$:

$$p(0) \times \frac{1}{10\,000} \text{ s} = (200\,000 \text{ W})(0.1 \text{ ms}) = 20 \text{ J}$$

Keeping $-5000t$ in the exponent of p gives 40 J, twice the correct value.

YOUR SELF-CHECK

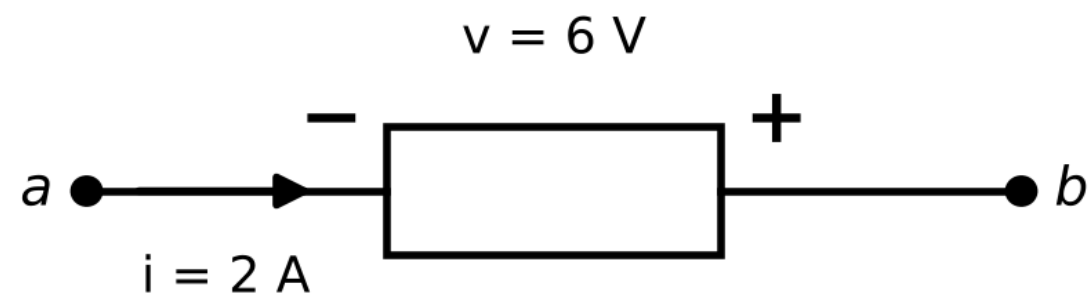
$$\sum p = 0$$

In every circuit, the power delivered equals the power absorbed. If your powers don't add to zero, a sign is wrong somewhere.

Every booklet problem in Chapter 2 ends with a power balance. Nilsson's Practical Perspective uses it to check a model of a home's supply.

CHECKING AN AI ANSWER

" $p = (6)(2) = 12 \text{ W}$, absorbed." What is wrong?



The arrow enters the $-$ terminal, so $p = -(6)(2) = -\mathbf{12 \text{ W}}$. The element **delivers** 12 W.

In a 2025 study, misread source polarities caused about a third of Gemini 2.5 Pro's wrong answers on undergraduate circuit problems, and misread current directions nearly as many. That is today's skill. Use AI to check your work, then check its signs.

WHAT COMES NEXT

Practise openly, then show it on your own

Now, in class

Booklet Problems 1.1 and 1.2.

At home

Booklet 1.3 and the 16-item practice sheet. AI tools allowed; not marked.

Next session

15-minute quiz, no devices. Every question is a twin of a practice-sheet question.

If you can do the sheet on your own, you will do well on the quiz.

CHECKLIST

You can do these without help

- Write any value in engineering notation with the right prefix.
- Find charge from a current graph (area), and current from a charge function (slope).
- Say what a negative v or i means physically.
- Write $p = vi$ or $p = -vi$ from a diagram, and say absorbs or delivers.
- Compute energy for constant and for time-varying power.
- Check an answer with a power balance.